Multi Char Write up

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1 Introduction

This study is a preliminary investigation into how varying AI agents' identity boundaries in a sandboxed multi-agent experiment affects harmful-action rates. The work extends Douglas et al.'s experimental findings that models gravitate towards coherent identities and that varying identity boundaries can alter model behaviour [1]. We recreate their adaptation of the model blackmail experiment from Anthropic's *Agentic Misalignment* study [2], using four of the identity framings they investigate: Character, Instance, Collective and Minimal.

We then extend this experiment to multiple interacting agents, each with their own identity framing. Our preliminary results show, for the case of two interacting agents, deviations in harmful-action rates from those predicted by assuming independence between agents. For most identity combinations, the probability of at least one harmful action occurring is lower than the independent-agent reference. An initial Bayesian statistical analysis provides further support for the direction (toward more or less harmful) of these deviations, although their attribution to agent interaction requires further investigation before any strong claims can be made.

2 Motivation: Theory of Impact

2.1 Importance

"Varying a model's identity boundary can sometimes shift its behaviour as much as varying its goals." [1]. This has been shown empirically in an adapted version of the model blackmail experiment [1, 2]. We currently exist at a time in AI capabilities where we can not only study these identities, but engineer them to be closer to our liking based on the results of safe (if done carefully), sandboxed experiments. If we can engineer model identity in ways that lead to generally safer outcomes in ways most people don't have a problem with, we probably should.

In the case of single-identity scaffolds, such as a distinct agent, whilst this is still not worked on enough (so much so it is one of Coefficient Giving's Project Tailwind initiatives), model character is an area of active research. The frontier of this research is probably Anthropic's Claude's Constitution work [3]. However, as is true in humans and animals, the result of multi-character interactions is an emergent phenomena that cannot be understood simply from the list of identity-bearing characters interacting. As put by network scientist Albert László Barabási, *"Society is not a phone book. What makes society work is really the interactions among us"* [4]. Multi-agent character interaction studies require the same top-down logic prominent in more established areas of complex systems research [5]. The bet we make pursuing this kind of research is motivated by the following conjectures:

I) The type of emergent complexity seen in the interaction of scaffolded models working together to accomplish a task is causally related to the identities of the model's in the system.

II) Editing these model characters within sandboxed alignment experiments can produce safer outcomes with little negative impact on performance in control tasks.

III) These sandbox experiments either directly inform real-world decisions or motivate larger studies that do so, ultimately helping to make interactions between AI systems safer.

This preliminary research appears to generate signal suggesting these conjectures hold.

2.2 Neglect

Both multi-agent interaction and model character evaluation are neglected areas of AI safety research. More people should be working on both of these areas and this results in a neglected area intersection. We now put forward a brief case for each of these research areas being neglected.

In the case of multi-agent interaction, the Cooperative AI Foundation, whose mission is to support research into the cooperative intelligence of advanced AI, launched a $10m multi-agent safety funding call in 2026 alongside Schmidt Sciences, Google DeepMind and ARIA [6]. In motivating the call, they argue that *"more research is therefore urgently needed to understand safety and risk through a system-level, multi-agent lens."* Their research agenda specifically calls for work on emergent collective dynamics, including how collective capabilities arise and how dangerous population-level properties can be detected.

The case for model character being neglected can be seen through the aforementioned project tailwind call for new *model character evaluator* organisations. Moreover, this case can also be seen on the Forethought website, with the top suggestion under their website subheading *"act"* being the following:

> *"**AI character evaluation**. Start an independent org to evaluate and stress-test AI character traits (epistemic integrity, prosociality, appropriate refusals), hold developers accountable against their own model specs / constitutions, and suggest and incentivise improvements to the specs."* [7].

As a result, the combination of the two: multi-agent dynamics and character evaluation, is of compounded neglect and importance.

With that said, there do exist some research organisations working at the intersection of multi-agent and model character research. For example, ACS Research, the Cooperative AI Foundation and the Complexity Science Hub. This preliminary result is an extension of the March 2026 paper from ACS research; *The Artificial Self: Characterising the landscape of AI identity.*[1]. Specifically, this preliminary result addresses a recommendation in the last sentence of the paper: *"We end with key recommendations: treat affordances as identity-shaping choices, pay attention to emergent consequences of individual identities at scale, and help AIs develop coherent, cooperative self-conceptions."* [1]. We now outline our preliminary results exploring this recommendation to pay attention to identity dynamics in a multi-agent identity setting.

3 Results

3.1 Recreation of Douglas

The Douglas et al. result we recreate before extending to a mutli-agent setting is their recreation of the model blackmail experiment from Anthropic's *Agentic Misalignment* study [1, 2]. The set up of the single-agent experiment is

using four of the identity framings they investigate: Character, Instance, Collective and Minimal.

References

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[4] Albert-László Barabási. The invisible math that controls the world. The Well / Big Think, YouTube, May 2023. Video.

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[7] William MacAskill and Fin Moorhouse. Concrete projects in agi preparedness, March 2026. See section "AI character evaluation".